oppgave 1

arbeidsfluid: Nitrogen (N_2)

innløp:
$T_1 = 310[k]$
$P_1 = 700[kPa]$

utløp:
$T_2 = 590[k]$
$P_2 = 700[kPa]$

omgivelser:
$T_0 = 300[k]$
$P_0 = 100[kPa]$

innstrømsvolum:
$\dot{V} = 904\left[\frac{m^3}{s}\right]$

gjennomsnittstemperatur
$T_{avg} = \frac{310+590}{2} = 450[k]$

fra tabell A-1 og A-2(b) ved 450k for N_2:

gasskonstant $(R) = 0,2968\left[\frac{kJ}{kg \cdot k}\right]$

spesifikk varmekapasitet $(c_p) = 1,049\left[\frac{kJ}{kg \cdot k}\right]$, $(c_v) = 0,752\left[\frac{kJ}{kg \cdot k}\right]$

isentropisk eksponent $(k) = 1,395$

a)

for et åpent, stasjonært og adiabatisk system ($\dot{Q} = 0$, $\Delta ke \approx 0$, $\Delta pe \approx 0$):

$$w_{komp,inn} = h_2 - h_1 = c_p \cdot (T_2 - T_1)$$

$$w_{komp,inn} = 1,049 \cdot (590 - 310) = \underline{\underline{293,72\left[\frac{kJ}{kg}\right]}}$$

b)

isentropisk utløsetemperatur (T_{2s}):

$$\frac{T_{2s}}{T_1} = \left(\frac{P_2}{P_1}\right)^{\frac{k-1}{k}} \Rightarrow T_{2s} = 310\left(\frac{700}{100}\right)^{\frac{1,395-1}{1,395}} = 310(1)^{0,28315} = 541,15[k]$$

isentropisk arbeid

$$w_{s,inn} = c_p(T_{2s} - T_1) = 1.049(541,15 - 310) = \underline{\underline{242,48\left[\frac{kJ}{kg}\right]}}$$

c)

$n = 1,2$

optimalt mellomtrykk:

$$p_x = \sqrt{P_1 P_2} = \sqrt{100 \cdot 700} = 264.58[kpa]$$

for 2-trinns polytropisk kompresjon:

$$w_{2\text{-trinn}} = 2\frac{n}{n-1}RT_1\left(\left(\frac{p_x}{p_1}\right)^{\frac{n-1}{n}} - 1\right)$$

$$= 12 \cdot 0,2968 \cdot 310 \cdot (1,17188 - 1) = \underline{189,75\left[kJ/kg\right]}$$

d)

$$\text{massestrøm}\,(\dot{m}) = \frac{p_1 \dot{V}_1}{RT_1} = \frac{100 \cdot 0,04}{0,2968 \cdot 310} = 0,043475\left[kg/s\right]$$

endring i spesifik entropi

$$\Delta s = c_p \ln\left(\frac{T_2}{T_1}\right) - R \ln\left(\frac{p_2}{p_1}\right) = 1,047 \cdot \ln\left(\frac{590}{310}\right) - 0,2968 \cdot \ln\left(\frac{700}{100}\right)$$

$$= 0,098\left[kJ/kgK\right]$$

entropigenereringsraten $(\dot{s}_{gen})$ siden $\dot{Q} = 0$:

$$\dot{s}_{gen} = \dot{m}\Delta s = 0,043475 \cdot 0,098 = \underline{0,00426\left[kW/K\right]}$$

ekserg i ødeleggelsesrate $\left(\dot{X}_{ødelagt}\right) = T_0 \cdot \dot{s}_{gen} = 300 \cdot 0,00426 = \underline{1,28\left[kW\right]}$

oppg 2

$\dot{m} = 5\ [kg]$

$\dot{Q}_{inn,kjel} = 16620[kW]$

$h_1 = 317{,}62\left[{}^{kJ}/_{kg\cdot k}\right]$ (40 [kPa] mettet veske, $v_1 = 0{,}001026\left[{}^{m^3}/_{kg}\right]$)

$h_2 = 526{,}62\left[{}^{kJ}/_{kg\cdot k}\right]$ (7 [MPa])

$h_3 = 3650{,}6\left[{}^{kJ}/_{kg\cdot k}\right]$, $s_3 = 7{,}3706\left[{}^{kJ}/_{kg\cdot k}\right]$ (7 MPa, 600°C)

$h_4 = 2636{,}1\left[{}^{kJ}/_{kg\cdot k}\right]$ (40 [kPa])

$T_{kjele} = 1000[°C] = 1273[K]$, $T_0 = 15[°C] = 289{,}15[K]$

a)

$$\dot{W}_{turbin} = \dot{m}(h_3 - h_4) = 5(3650{,}6 - 2636{,}1) = 5072{,}5\,[kW] \approx 5{,}07\,[MW]$$

b)

$$w_{p,inn} = h_2 - h_1 = 526{,}62 - 317{,}62 = 9\left[{}^{kJ}/_{kg}\right]$$

$$w_{s,p} = v_1(P_2 - P_1) = 0{,}001026(7000 - 40) = 7{,}141\left[{}^{kJ}/_{kg}\right]$$

$$\eta_p = \frac{w_{s,p}}{w_{p,inn}} = \frac{7{,}141}{9} = 0{,}7935 \approx (79{,}4\%)$$

c)

$$s_4 = s_3 = 7,3706 \left[{}^{kJ}/_{kg \cdot K} \right]$$

$$s_f = 1,0261$$

$$s_g = 7,6691$$

$$x_{4s} = \frac{s_4 - s_f}{s_g - s_f} = \frac{7,3706 - 1,0261}{7,6691 - 1,0261} = 0,95507$$

$$h_{fg} = 2318,4$$

$$h_{4s} = h_1 + x_{4s} \cdot h_{fg} = 317,62 + 0,95507 \cdot 2318,4 = 2531,82 \left[{}^{kJ}/_{kg} \right]$$

$$\eta_t = \frac{h_3 - h_4}{h_3 - h_{4s}} = \frac{3650,6 - 2636,1}{3650,6 - 2531,82} = \frac{1014,5}{1118,78} = 0,9068 \approx (90,7\%)$$

d) $s_4 = s_g = 7,6691 \left[{}^{kJ}/_{kg \cdot K} \right]$

$$\eta_{2,turbin} = \frac{h_3 - h_4}{(h_3 - T_0 \cdot s_3) - (h_4 - T_0 \cdot s_4)} = \frac{1014,5}{(3650,6 - 288,15 \cdot 7,3706) - (2636,1 - 288,15 \cdot 7,6691)}$$

$$= \frac{1014,5}{1160,69} = 0,9217 \approx (92,2\%)$$

e)

$$\dot{W}_{pumpe} = \dot{m}(h_2 - h_1) = 5 \cdot 9 = 45 \left[kW \right]$$

$$\dot{W}_{netto} = \dot{W}_{turbin} - \dot{W}_{pumpe} = 5072,5 - 45 = 5027,5 \left[kW \right]$$

$$\dot{X}_{inn} = \dot{Q}_{inn} \left(1 - \frac{T_0}{T_{kilde}} \right) = 1620 \left(1 - \frac{288,15}{1273,15} \right) = 1258,4 \left[kW \right]$$

⇓

$$\eta_{2,anlegg} = \frac{\dot{W}_{netto}}{\dot{X}_{inn}} = \frac{5027,5}{12858,4} = \underline{\underline{0,391 = (39,1\%)}}$$

oppgave 3

$P_1 = 200\,[kPa]$, $T_1 = 20\,[°C]$, $\dot{m}_1 = 100\,[kg/min] \approx 1,6667\,[kg/s]$

$P_2 = 200\,[kPa]$, $T_2 = 200\,[°C]$

$P_3 = 200\,[kPa]$, $T_3 = 90\,[°C]$

$T_0 = 20\,[°C]$, $P_0 = 100\,[kPa]$

$h_1 = 83,91\,[kJ/kg]$, $s_1 = 0,2965\,[kJ/kg \cdot k]$

$h_2 = 2870,7\,[kJ/kg]$, $s_2 = 7,5081\,[kJ/kg \cdot k]$

$h_3 = 377,04\,[kJ/kg]$, $s_3 = 1,1929\,[kJ/kg \cdot k]$

$h_0 = 83,91\,[kJ/kg]$, $s_0 = 0,2965\,[kJ/kg \cdot k]$

a)

$$\dot{m}_1(h_1 - h_3) + \dot{m}_2(h_2 - h_3) = 0$$

$$\dot{m}_2 = \dot{m}_1 \frac{h_3 - h_1}{h_2 - h_3} = 1,6667 \frac{377,04 - 83,91}{2870,7 - 377,04} = \underline{\underline{0,1957\,[kg/s]}}$$

b)

$$\dot{m}_3 = 1,6667 + 0,1959 = 1,8626 \left[kg/s \right]$$

$$\dot{S}_{gen} = \dot{m}_3 s_3 - \left(\dot{m}_1 s_1 + \dot{m}_2 s_2 \right)$$

$$\dot{S}_{gen} = 1,8626 \cdot 1,1929 - (1,6667 \cdot 0,2965 + 0,1959 \cdot 7.5081) = 2,2219 - 1,9651 = 0,2568 \left[kW/k \right]$$

$$\dot{X}_{podlagt} = T_0 \dot{S}_{gen} = 293,15 \cdot 0,2568 = 75,27 [kW]$$

c)

$$\dot{x}_3 = (h_3 - h_0) - T_0 (s_3 - s_0) = (377,04 - 83,91) - 293,15 (1,1929 - 0,2965)$$

$$\dot{x}_3 = 30,35 \left[kJ/kg \right]$$

$$\dot{X}_3 = \dot{m}_3 x_3 = 1,8626 \cdot 30,35 = 56,55 [kW]$$